

Ejercicio Clase 9

Consigna: Por cada ejercicio, escribir el código y agregar una captura de pantalla del resultado obtenido.

1. Crear una base de datos en Hive llamada northwind_analytics

```
hive> create database northwind_analytics;  
OK  
Time taken: 0.435 seconds
```

2. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de clientes junto a la cantidad de productos vendidos ordenados de mayor a menor (campos customer_id, company_name, productos_vendidos). Luego ingestar el archivo a HDFS (carpeta /scoop/ingest/clientes). Pasar la password en un archivo

Con respecto a la password, Gemini me recomendó guardar el archivo en una carpeta llamada ".scoop_password". El punto al principio hace que el directorio se mantenga "oculto". Los pasos de creación de carpeta, archivo y permisos:

```
hadoop@c210c2373003:/$ mkdir /home/hadoop/.scoop_passwords  
hadoop@c210c2373003:/$ echo -n "edvai" > /home/hadoop/.scoop_passwords/  
db_password.txt  
hadoop@c210c2373003:/$ chmod 400 /home/hadoop/.scoop_passwords/db_passw  
ord.txt
```

Primero quería armar la sentencia de scoop así:

```
scoop import \  
-connect jdbc:postgresql://172.17.0.3:5432/northwind \  
-username postgres \  
-query "SELECT c.customer_id, c.company_name, SUM(od.quantity) AS  
productos_vendidos FROM customers c INNER JOIN orders o ON c.customer_id =  
o.customer_id INNER JOIN order_details od ON o.order_id = od.order_id GROUP BY  
c.customer_id, c.company_name order by productos_vendidos desc WHERE  
\$CONDITIONS" \  
-m 1 \  
-password-file file:///home/hadoop/.scoop_passwords/db_password.txt \  
-target-dir /scoop/ingest/clientes \  
-as-parquetfile \  
-delete-target-dir
```

Pero no me tomaba el WHERE\\$CONDITIONS y también tenía problemas para llamar al archivo de password que no lo encontraba.

In SQL, the `WHERE` clause must come before the `GROUP BY` clause. Since you have a `GROUP BY` clause, you cannot place the `WHERE $CONDITIONS` (which becomes `WHERE (1 = 0)` during the initial metadata check) after it.

✓ Solution

To fix this, you must wrap your entire query, including the `GROUP BY` clause, in an **outer** `SELECT` statement and place the `\$CONDITIONS` token in the **outer** `WHERE` clause. This is the standard way to use `--query` with aggregation in Sqoop.

Luego de algunas pruebas y algo de IA, la query quedó como sub-query y la password le saqué los asteriscos al inicio y final y funcionó.

Script ingest clientes.sh:

```
sqoop import \  
  --connect jdbc:postgresql://172.17.0.3:5432/northwind \  
  --username postgres \  
  --query "SELECT sub.* FROM (SELECT c.customer_id, c.company_name,  
SUM(od.quantity) AS productos_vendidos FROM customers c INNER JOIN orders o ON  
c.customer_id = o.customer_id INNER JOIN order_details od ON o.order_id = od.order_id  
GROUP BY c.customer_id, c.company_name order by productos_vendidos desc) AS sub  
WHERE \$CONDITIONS" \  
  -m 1 \  
  --password-file file:///home/hadoop/.sqoop_passwords/db_password.txt \  
  --target-dir /sqoop/ingest/clientes \  
  --as-parquetfile \  
  --delete-target-dir
```

lectura parquet en pyspark:

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000//sqoop/ingest/clientes")  
>>> df.show(10)  
+-----+-----+-----+  
|customer_id|      company_name|productos_vendidos|  
+-----+-----+-----+  
|      SAVEA| Save-a-lot Markets|          4958|  
|      ERNSH| Ernst Handel|          4543|  
|      QUICK| QUICK-Stop|          3961|  
|      HUNGO| Hungry Owl All-Ni...|          1684|  
|      FRANK| Frankenversand|          1525|  
|      RATTC| Rattlesnake Canyo...|          1383|  
|      FOLKO| Folk och fä HB|          1234|  
|      HILAA| HILARION-Abastos|          1096|  
|      SUPRD| Suprêmes délices|          1072|  
|      WHITC| White Clover Markets|          1063|  
+-----+-----+-----+  
only showing top 10 rows
```

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000//sqoop/ingest/clientes")
>>> df.count()
89
```

Cantidad de registros: 89.

Archivos parquet en HDFS:

```
hadoop@c210c2373003:~$ hdfs dfs -ls /sqoop/ingest/clientes
Found 3 items
drwxr-xr-x - hadoop supergroup          0 2025-10-28 12:25 /sqoop/ingest/clientes/.metadata
drwxr-xr-x - hadoop supergroup          0 2025-10-28 12:25 /sqoop/ingest/clientes/.signals
-rw-r--r-- 1 hadoop supergroup    3845 2025-10-28 12:25 /sqoop/ingest/clientes/f26b3e09-9725-4463-a924-396705dbd71e.parquet
hadoop@c210c2373003:~$
```

3. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de órdenes junto a qué empresa realizó cada pedido (campos order_id, shipped_date, company_name, phone). Luego ingestar el archivo a HDFS (carpeta /sqoop/ingest/envíos). Pasar la password en un archivo

Script ingest_envios.sh :

```
sqoop import \
--connect jdbc:postgresql://172.17.0.3:5432/northwind \
--username postgres \
--query "select o.order_id , TO_CHAR(o.shipped_date, 'YYYY-MM-DD') AS shipped_date ,
c.company_name , c.phone from orders o inner join customers c on o.customer_id =
c.customer_id WHERE \${CONDITIONS}" \
-m 1 \
--password-file file:///home/hadoop/.sqoop_passwords/db_password.txt \
--target-dir /sqoop/ingest/envios \
--as-parquetfile \
--delete-target-dir
```

lectura parquet en pyspark:

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000//sqoop/ingest/envios")
>>> df.show(10)
+-----+-----+-----+-----+
|order_id|shipped_date|company_name|phone|
+-----+-----+-----+-----+
| 10248| 1996-07-16|Vins et alcools C...| 26.47.15.10|
| 10249| 1996-07-10|Toms Spezialitäten| 0251-031259|
| 10250| 1996-07-12|Hanari Carnes|(21) 555-0091|
| 10251| 1996-07-15|Victuailles en stock| 78.32.54.86|
| 10252| 1996-07-11|Suprêmes délices|(071) 23 67 22 20|
| 10253| 1996-07-16|Hanari Carnes|(21) 555-0091|
| 10254| 1996-07-23|Chop-suey Chinese| 0452-076545|
| 10255| 1996-07-15|Richter Supermarkt| 0897-034214|
| 10256| 1996-07-17|Wellington Import...|(14) 555-8122|
| 10257| 1996-07-22|HILARION-Abastos|(5) 555-1340|
+-----+-----+-----+-----+
only showing top 10 rows
```

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000//sqoop/ingest/envios")
>>> df.count()
830
```

Cantidad de registros: 830.

Archivo parquet en hdfs:

```
hadoop@c210c2373003:~/scripts$ hdfs dfs -ls /sqoop/ingest/envios
Found 3 items
drwxr-xr-x - hadoop supergroup          0 2025-10-28 19:37 /sqoop/ingest/envios/.metadata
drwxr-xr-x - hadoop supergroup          0 2025-10-28 19:37 /sqoop/ingest/envios/.signals
-rw-r--r-- 1 hadoop supergroup    25328 2025-10-28 19:37 /sqoop/ingest/envios/0c648f23-5647-403b-80b8-ef9f2a73bd0
c.parquet
```

4. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de detalles de órdenes (campos order_id, unit_price, quantity, discount). Luego ingestar el archivo a HDFS (carpeta /sqoop/ingest/order_details). Pasar la password en un archivo

Script ingest_order_details.sh:

```
sqoop import \
--connect jdbc:postgresql://172.17.0.3:5432/northwind \
--username postgres \
--query "select order_id, unit_price, quantity, discount from order_details WHERE
\${CONDITIONS}" \
-m 1 \
--password-file file:///home/hadoop/.sqoop_passwords/db_password.txt \
--target-dir /sqoop/ingest/order_details \
--as-parquetfile \
--delete-target-dir
```

En spark:

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000//sqoop/ingest/order_details")
>>> df.show(10)
+-----+-----+-----+-----+
|order_id|unit_price|quantity|discount|
+-----+-----+-----+-----+
|  10248 |    14.0 |      12 |    0.0 |
|  10248 |     9.8 |      10 |    0.0 |
|  10248 |    34.8 |       5 |    0.0 |
|  10249 |    18.6 |       9 |    0.0 |
|  10249 |    42.4 |      40 |    0.0 |
|  10250 |     7.7 |      10 |    0.0 |
|  10250 |    42.4 |      35 |    0.15|
|  10250 |    16.8 |      15 |    0.15|
|  10251 |    16.8 |       6 |    0.05|
|  10251 |    15.6 |      15 |    0.05|
+-----+-----+-----+-----+
```

```
>>> df = spark.read.parquet("hdfs://172.17.0.2:9000/sqoop/ingest/order_details")
>>> df.count()
2155
```

Cantidad de registros: 2155.

Archivo parquet en HDFS:

```
hadoop@c210c2373003:~/scripts$ hdfs dfs -ls /sqoop/ingest/order_details
Found 3 items
drwxr-xr-x - hadoop supergroup 0 2025-10-28 19:37 /sqoop/ingest/order_details/.metadata
drwxr-xr-x - hadoop supergroup 0 2025-10-28 19:37 /sqoop/ingest/order_details/.signals
-rw-r--r-- 1 hadoop supergroup 12292 2025-10-28 19:37 /sqoop/ingest/order_details/d8b0dada-a31c-4b5c-bd76-a0b71a6800b2.parquet
```

5. Generar un archivo .py que permita mediante Spark insertar en hive en la db northwind_analytics en la tabla products_sold, los datos del punto 2, pero solamente aquellas compañías en las que la cantidad de productos vendidos fue mayor al promedio.

Script `t_products_sold.py`:

```
from pyspark.context import SparkContext
from pyspark.sql.session import SparkSession
sc = SparkContext("local")
spark = SparkSession(sc)

from pyspark.sql import HiveContext
hc = HiveContext(sc)

##Leyendo archivo clientes
df = spark.read.parquet("hdfs://172.17.0.2:9000/sqoop/ingest/clientes")

#Crea las vista del parquet
df.createOrReplaceTempView("cliente_vista_1")

#Filtro compañías que vendieron más que el promedio de ventas global

df_01_filtro = spark.sql("WITH T_Ventas AS (SELECT customer_id, company_name,
productos_vendidos FROM cliente_vista_1), T_ConPromedio AS (SELECT *,
AVG(productos_vendidos) OVER () AS promedio_global_ventas FROM T_Ventas) SELECT
customer_id, company_name, productos_vendidos, promedio_global_ventas FROM
T_ConPromedio WHERE productos_vendidos > promedio_global_ventas ")

#Hago la vista para insertar
df_01_filtro.createOrReplaceTempView("compania_insertar")

#Hace load en HIVE
hc.sql("insert into northwind_analytics.products_sold select * from compania_insertar")
```

La tabla con el promedio global de ventas y las compañías que vendieron más que el promedio. Las 5 primeras filas:

customer_id	company_name	productos_vendidos	promedio_global_ventas
SAVEA	Save-a-lot Markets	4958	576.5955056179776
ERNSH	Ernst Handel	4543	576.5955056179776
QUICK	QUICK-Stop	3961	576.5955056179776
HUNGO	Hungry Owl All-Ni...	1684	576.5955056179776
FRANK	Frankenversand	1525	576.5955056179776

Las 5 últimas filas:

```
>>> df_01_filtro.orderBy("productos_vendidos", ascending=True).limit(5).show()
```

customer_id	company_name	productos_vendidos	promedio_global_ventas
OLDWO	Old World Delicat...	603	576.5955056179776
PICCO	Piccolo und mehr	624	576.5955056179776
OTTIK	Ottilies Käseladen	639	576.5955056179776
AROUT	Around the Horn	650	576.5955056179776
RICAR	Ricardo Adocicados	660	576.5955056179776

6. Generar un archivo .py que permita mediante Spark insertar en hive en la tabla products_sent, los datos del punto 3 y 4, de manera tal que se vean las columnas order_id, shipped_date, company_name, phone, unit_price_discount (unit_price with discount), quantity, total_price (unit_price_discount * quantity). Solo de aquellos pedidos que hayan tenido descuento.

Script t_products_sent.py:

```
from pyspark.context import SparkContext
from pyspark.sql.session import SparkSession
sc = SparkContext('local')
spark = SparkSession(sc)
```

```
from pyspark.sql import HiveContext
hc = HiveContext(sc)
```

```
##Leyendo archivos de envios y order_details
df_envios = spark.read.parquet("hdfs://172.17.0.2:9000/sqoop/ingest/envios")
df_od = spark.read.parquet("hdfs://172.17.0.2:9000/sqoop/ingest/order_details")
```

```
#Creando las vista de cada parquet
df_envios.createOrReplaceTempView("envios_vista")
df_od.createOrReplaceTempView("od_vista")
```

#Queries

```
df_filtro = spark.sql("WITH Detalles_Calculados AS (SELECT od.order_id, od.quantity, (od.unit_price * (1 - od.discount)) AS unit_price_discount FROM od_vista od where od.discount > 0 ) select dc.order_id, cast(o.shipped_date as date), o.company_name, o.phone, dc.unit_price_discount, dc.quantity, (dc.unit_price_discount * dc.quantity) AS total_price from Detalles_Calculados dc INNER join envios_vista o ON dc.order_id = o.order_id ;")
```

#Hago la vista para insertar

```
df_filtro.createOrReplaceTempView("vista_insertar")
```

#Hace load en HIVE

```
hc.sql("insert into northwind_analytics.products_sent select * from vista_insertar")
```

La tabla solo con aquellos pedidos con descuento, los campos unit_price_discount y total_price:

```
>>> df_filtro = spark.sql("WITH Detalles_Calculados AS (SELECT od.order_id, od.quantity, (od.unit_price * (1 - od.discount)) AS unit_price_discount FROM od_vista od where od.discount > 0 ) select dc.order_id, cast(o.shipped_date as date), o.company_name, o.phone, dc.unit_price_discount, dc.quantity, (dc.unit_price_discount * dc.quantity) AS total_price from Detalles_Calculados dc INNER join envios_vista o ON dc.order_id = o.order_id ;")
>>> df_filtro.show(10)
```

order_id	shipped_date	company_name	phone	unit_price_discount	quantity	total_price
10250	1996-07-12	Hanari Carnes	(21) 555-0091	14.28	15	214.2
10250	1996-07-12	Hanari Carnes	(21) 555-0091	36.04	35	1261.4
10251	1996-07-15	Victuailles en stock	78.32.54.86	14.820001	15	222.3
10251	1996-07-15	Victuailles en stock	78.32.54.86	15.959999	6	95.759995
10252	1996-07-11	Suprêmes délices	(071) 23 67 22 20	1.9	25	47.5
10252	1996-07-11	Suprêmes délices	(071) 23 67 22 20	61.56	40	2462.4001
10254	1996-07-23	Chop-suey Chinese	0452-076545	16.320002	21	342.72003

7. Realizar un proceso automático en Airflow que orqueste los pipelines creados en los puntos anteriores. Crear un grupo para la etapa de ingest y otro para la etapa de process. Correrlo y mostrar una captura de pantalla (del DAG y del resultado en la base de datos)

DAG:

```
from datetime import timedelta
from airflow import DAG
from airflow.operators.bash import BashOperator
from airflow.operators.dummy import DummyOperator
from airflow.utils.dates import days_ago
from airflow.utils.task_group import TaskGroup
from airflow.models.baseoperator import chain
```

```
args = {
    'owner': 'airflow',
}
```

```
with DAG (
```

```

dag_id='ingest_clase9',
default_args=args,
schedule_interval='0 0 * * *',
start_date=days_ago(2),
dagrun_timeout=timedelta(minutes=60),
tags=['ingest', 'transform', 'sqoop'],
params={"example_key": "example_value"},
) as dag:

    inicio_proceso = DummyOperator(
        task_id="Inicio_proceso",
    )

    finaliza_proceso = DummyOperator(
        task_id='Finaliza_proceso',
    )

    # CREACION DEL GRUPO DE TAREAS 'INGEST'
    with TaskGroup("Ingest", tooltip="Tareas de Extracción de Datos") as
ingest_group:

        export_table_1 = BashOperator(
            task_id='export_table_1',
            bash_command='/usr/bin/sh /home/hadoop/scripts/ingest_clientes.sh
',
        )

        export_table_2 = BashOperator(
            task_id='export_table_2',
            bash_command='/usr/bin/sh /home/hadoop/scripts/ingest_envios.sh
',
        )

        export_table_3 = BashOperator(
            task_id='export_table_3',
            bash_command='/usr/bin/sh
/home/hadoop/scripts/ingest_order_details.sh ',
        )

    # CREACION DEL GRUPO DE TAREAS 'PROCESS'
    with TaskGroup("Process", tooltip="Tareas de Procesamiento de Datos") as
process_group:

        processing_table_products_sold = BashOperator(
            task_id='processing_table_products_sold',
            bash_command='ssh hadoop@172.17.0.2
/home/hadoop/spark/bin/spark-submit --files
/home/hadoop/hive/conf/hive-site.xml /home/hadoop/scripts/t_products_sold.py
',
        )

        processing_table_products_sent = BashOperator(

```



```

        task_id='processing_table_products_sent',
        bash_command='ssh hadoop@172.17.0.2
/home/hadoop/spark/bin/spark-submit --files
/home/hadoop/hive/conf/hive-site.xml /home/hadoop/scripts/t_products_sent.py
',
    )

```

```

# La tarea de inicio es seguida por el grupo completo 'ingest'
inicio_proceso >> ingest_group

```

```

# El grupo completo 'ingest' es seguido por el grupo completo 'Process'
ingest_group >> process_group

```

```

# El grupo completo 'Process' es seguido por la tarea final
process_group >> finaliza_proceso

```

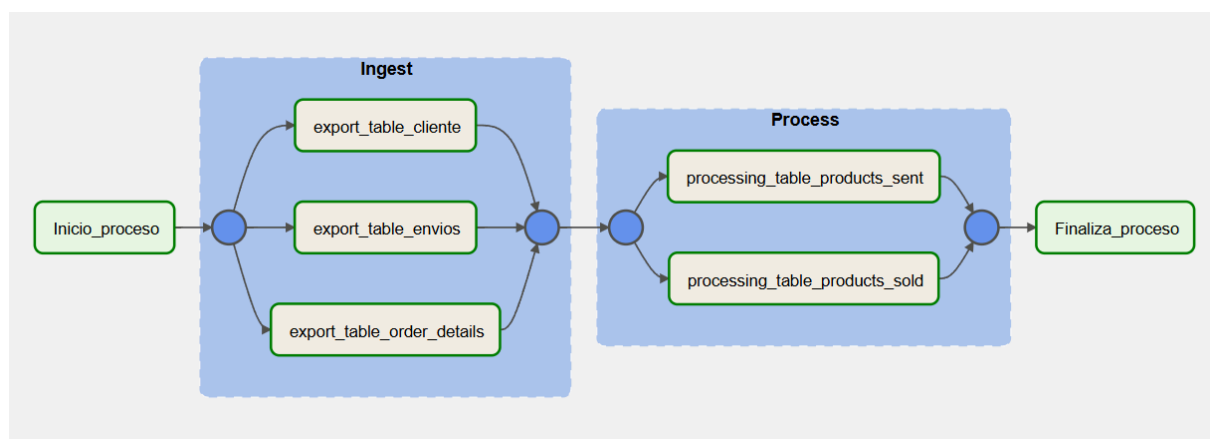
```

if __name__ == "__main__":
    dag.cli()

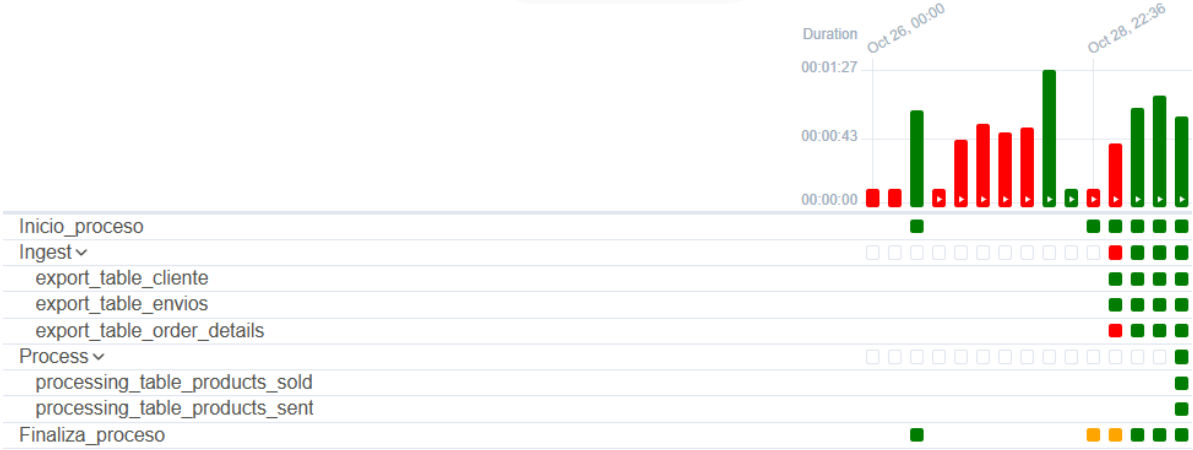
```

DAGs

All 1		Active 1	Paused 0	* sqoop			
DAG	Owner	Runs	Schedule	Last Run	Next Run		
ingest_clase9 ingest sqoop transform	airflow	6 9	00***	2025-10-29, 00:58:29	2025-10-29, 00:00:00		



XComs



DAG
ingest_clase9

DAG Details

DAG Runs Summary

Total Runs Displayed	15
■ Total success	6
■ Total failed	9
First Run Start	2025-10-28, 21:51:50 UTC
Last Run Start	2025-10-29, 00:58:31 UTC
Max Run Duration	00:01:27
Mean Run Duration	00:00:39
Min Run Duration	00:00:03

DAG Summary

Total Tasks	7
DummyOperators	2
BashOperators	5

Tabla products_sold:

```
hive> describe products_sold;
OK
customer_id      string
company_name     string
productos_vendidos bigint
promedio_global_ventas float
Time taken: 0.038 seconds, Fetched: 4 row(s)
```

```
hive> select * from products_sold order by productos_vendidos limit 10;
```

MapReduce Jobs Launched:

Stage-Stage-1: HDFS Read: 682008 HDFS Write: 589 SUCCESS

Total MapReduce CPU Time Spent: 0 msec

OK

OLDWO	Old World Delicatessen	603	576.5955
PICCO	Piccolo und mehr	624	576.5955
OTTIK	Ottilies Käseladen	639	576.5955
AROUT	Around the Horn	650	576.5955
RICAR	Ricardo Adocicados	660	576.5955
BLONP	Blondesddsl père et fils	666	576.5955
WARTH	Wartian Herkku	737	576.5955
VAFFE	Vaffeljernet	792	576.5955
LEHMS	Lehmanns Marktstand	794	576.5955
RICSU	Richter Supermarkt	810	576.5955

Time taken: 1.747 seconds, Fetched: 10 row(s)

Tabla products_sent:

```
hive> describe products_sent;
```

OK

order_od	int
shipped_date	date
company_name	string
phone	string
unit_price_discount	float
quantity	int
total_price	float

Time taken: 0.032 seconds, Fetched: 7 row(s)

```
hive> select * from products_sent order by shipped_date desc limit 10;
```

MapReduce Jobs Launched:

Stage-Stage-1: HDFS Read: 937140 HDFS Write: 8868 SUCCESS

Total MapReduce CPU Time Spent: 0 msec

OK

11063	1998-05-06	Hungry Owl All-Night Grocers	2967	542	16.56	40	662.39996
11063	1998-05-06	Hungry Owl All-Night Grocers	2967	542	8.684999	30	260.55
11050	1998-05-05	Folk och fä HB 0695-34 67 21	16.199999	50	809.99994		
11064	1998-05-04	Save-a-lot Markets (208) 555-8097	29.519999	25	737.99994		
11049	1998-05-04	Gourmet Lanchonetes (11) 555-9482	30.4	4	121.6		
11064	1998-05-04	Save-a-lot Markets (208) 555-8097	35.1	77	2702.7		
11049	1998-05-04	Gourmet Lanchonetes (11) 555-9482	15.2	10	152.0		
11064	1998-05-04	Save-a-lot Markets (208) 555-8097	21.599998	4	86.399994		
11047	1998-05-01	Eastern Connection (171) 555-0297	16.0125	30	480.37503		
11047	1998-05-01	Eastern Connection (171) 555-0297	13.5	25	337.5		

Time taken: 1.198 seconds, Fetched: 10 row(s)